

RETAIL INTELLIGENCE

Customer Shopping Behavior Analysis

Uncovering Spending Patterns • Customer Segments • Product Preferences

Dataset: 3,900 Transactions | 18 Feature Columns | 4 Product Categories

3,900

Total Transactions

\$59.76

Avg. Purchase Amount

3.75 / 5

Avg. Review Rating

01 Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover actionable insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

02 Dataset Summary

Total Rows	3,900
Total Columns	18
Demographics	Age, Gender, Location, Subscription Status
Purchase Details	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping Behavior	Discount Applied, Previous Purchases, Frequency, Review Rating, Shipping Type
Missing Data	37 missing values in Review Rating column — imputed via category median

03 Exploratory Data Analysis (Python)

Data preparation and cleaning was performed in Python using the pandas library. The following pipeline was applied:

- ▶ **Data Loading** – Imported dataset using pandas
- ▶ **Initial Exploration** – Used `df.info()` and `.describe()` to inspect structure and summary statistics
- ▶ **Missing Data Handling** – Imputed 37 missing Review Rating values with the category-level median
- ▶ **Column Standardization** – Renamed all columns to snake_case for consistency and documentation
- ▶ **Feature Engineering** – Engineered `age_group` by binning ages; created `purchase_frequency_days`
- ▶ **Data Consistency Check** – Verified that `discount_applied` and `promo_code_used` were redundant — dropped `promo_code_used`
- ▶ **Database Integration** – Connected Python script to PostgreSQL and loaded the cleaned DataFrame for SQL analysis

04 Data Analysis — SQL (PostgreSQL)

Ten structured queries were run against the PostgreSQL database to answer key business questions. Key findings are summarized below.

4.1 Revenue by Gender

Gender	Total Revenue (USD)
Female	\$75,191
Male	\$157,890

4.2 Top 5 Products by Average Rating

Rank	Product	Avg. Review Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80

Rank	Product	Avg. Review Rating
5	Skirt	3.78

4.3 Shipping Type — Average Purchase Amount

Shipping Type	Avg. Purchase (USD)
Express	\$60.48
Standard	\$58.46

4.4 Subscribers vs. Non-Subscribers

Status	Total Customers	Avg. Spend	Total Revenue
Subscriber	1,053	\$59.49	\$62,645
Non-Subscriber	2,847	\$59.87	\$170,436

4.5 Discount-Dependent Products (Top 5)

Product	Discount Rate (%)
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

4.6 Customer Segmentation

Segment	Number of Customers
Loyal	3,116
Returning	701
New	83

4.7 Top 3 Products per Category

Category	Rank 1	Rank 2	Rank 3
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	—

4.8 Revenue by Age Group

Age Group	Total Revenue (USD)
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

05 Interactive Dashboard — Power BI

An interactive Power BI dashboard was built to present all findings visually. The dashboard includes slicers for Subscription Status, Gender, Category, and Shipping Type, and displays the following panels:

- ▶ KPI cards: 3.9K customers, \$59.76 avg. purchase, 3.75 avg. rating
- ▶ Subscription status breakdown: 27% subscribers vs. 73% non-subscribers
- ▶ Revenue and sales volume by product category
- ▶ Revenue and sales volume by age group (horizontal bar charts)

06 Business Recommendations

▶ Boost Subscriptions

Promote exclusive perks and early-access offers to convert the 73% non-subscriber base.

▶ Customer Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment (currently 3,116 customers).

► **Review Discount Policy**

Hat, Sneakers and Coat have ~50% discount rates — balance sales boosts against margin erosion.

► **Product Positioning**

Highlight Gloves, Sandals, and Boots in marketing campaigns — all rated above 3.80.

► **Targeted Marketing**

Focus acquisition efforts on Young Adults (\$62,143 revenue) and Express-shipping customers (\$60.48 avg. spend), who represent the highest-value segments.

Prepared using Python (pandas, PostgreSQL) and visualized in Power BI • All data is from the retail transactions dataset